

Fourier decay for self-conformal measures

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¹Based on joint work with Simon Baker, <https://arxiv.org/abs/2406.12821> and Han Yu

Fourier transform of measures

- **Fourier transform** of a measure μ on $\mathbb{R}^k$ is the function $\widehat{\mu}: \mathbb{R}^k \rightarrow \mathbb{C}$,

$$\widehat{\mu}(\xi) = \int_{\mathbb{R}^k} e^{-2\pi i \langle \xi, x \rangle} d\mu(x).$$

- Questions:
 - Is μ Rajchman ($|\widehat{\mu}(\xi)| \rightarrow 0$ as $|\xi| \rightarrow \infty$)?
 - Does it have **polynomial Fourier decay**

$$|\widehat{\mu}(\xi)| \lesssim |\xi|^{-\varepsilon}?$$

- Riemann–Lebesgue lemma: absolutely continuous measures are Rajchman.
- Applications of polynomial Fourier decay: fractal uncertainty principles, Fourier restriction, Fourier uniqueness, normality of μ -typical points...

Self-similar and self-conformal measures

- Let $\Phi = \{S_i: D \rightarrow D\}_{i \in I}$ be an IFS of **conformal** contractions on a closed $D \subseteq \mathbb{R}^k$. Assume not all maps share a common fixed point.
- Given **strictly positive** weights $\{p_i\}_{i \in I}$, $\sum_i p_i = 1$, the **self-conformal measure** satisfies

$$\mu(A) = \sum_{i \in I} p_i \mu(S_i^{-1}(A)).$$

- The $(1/2, 1/2)$ measure on the middle-third Cantor set is not Rajchman.
- Erdős ('40), Kahane ('71), Solomyak ('22): Almost every self-similar measure has polynomial Fourier decay (though very few examples are known).

Nonlinear IFSs

Often **nonlinearity** in conformal IFS $\rightarrow$ polynomial Fourier decay.
Bourgain–Dyatlov ('17), Sahlsten–Stevens ('24) assuming separation,
using additive combinatorics.

Theorem 1

Consider an IFS of analytic contractions $\Phi = \{S_i: [0, 1] \rightarrow [0, 1]\}_i$ and assume there is i such that S_i is not affine. Then every self-conformal measure has polynomial Fourier decay.

No separation assumptions needed.

Proof of Theorem 1

- Proof splits depending on whether Φ is **conjugate to linear**, i.e. $\Phi = \{F \circ S_i \circ F^{-1}\}_i$ for an analytic diffeomorphism F and affine S_i .
- Non-conjugate case (Algom–Rodriguez Hertz–Wang ('23+), Baker–Sahlsten ('23+)):
Key step: proving a spectral gap type result for a family of transfer operators.
- Conjugate case (Algom–Chang–Wu–Wu (24+), Baker–B. (24+)):
Then μ is the **pushforward** of a self-similar measure under F .
Generalises Kaufman ('84), Mosquera–Shmerkin ('18) who assumed equal contractions.

Self-conformal measures on $\mathbb{C}$

Obstructions to Fourier decay in higher dimensions:

- Reducible measures (supported on a line) are non-Rajchman
- The pushforward of $\mu_{1/3}$ on $\mathbb{R}$ under $F(z) = z + iz^2$ is non-Rajchman

theorem 2

Consider an IFS of holomorphic conformal contractions $\Phi = \{S_i: D \rightarrow D\}_i$, assume some S_i is non-affine, and assume no analytic curve is preserved by all the maps. Then every self-conformal measure has polynomial Fourier decay.

- Non-conjugate case proved by Algom–Rodriguez Hertz–Wang ('24+).
- Conjugate case proved by B.–Yu ('24+)
- $k > 2$ difficult (rotations don't commute...)

Conjugate case

A self-similar IFS $\{x \mapsto r_i O_i + t_i\}_{i \in I}$ on $\mathbb{R}^k$ is **non-expanding** if $\forall \varepsilon > 0$,

$$\#\{O_{i_1} \circ \dots \circ O_{i_n} : i_1, \dots, i_n \in I\} \lesssim e^{\varepsilon n}.$$

Holds automatically if $k \in \{1, 2\}$.

theorem 3 (B.–Yu), implies conjugate case of Thms 1 & 2

Let μ be an irreducible non-expanding self-similar measure on $\mathbb{R}^k$.

Let U be an open neighbourhood of $\text{supp}(\mu)$.

Let $f: U \rightarrow \mathbb{R}^d$ be analytic; assume the graph of f is not contained in a proper affine subspace of $\mathbb{R}^{k+d}$.

Then $f_*\mu$ has polynomial Fourier decay.

- Decay can sometimes be quantified.
- We conjecture ‘non-expanding’ assumption can be removed.

Polynomial Fourier decay on average (PFDA)

- To prove theorem 3, need to show the **self-similar measure** μ has **PFDA**: $\forall \varepsilon \exists \delta$ s.t.
 $\text{Leb}(\{\xi \in \mathbb{R}^k : |\widehat{\mu}(\xi)| \geq R^{-\delta}, |\xi| < R\}) \lesssim R^{-\varepsilon}.$
- Baker–Khalil–Sahlsten ('24+): if μ has PFDA and is self-similar with some contraction ratios r, r' s.t. $\log r / \log r'$ is Diophantine then $|\widehat{\mu}(\xi)| \lesssim (\log |\xi|)^{-\varepsilon}.$

theorem

A self-similar measure has PFDA if and only if it is irreducible.

- Forward implication is trivial.
- Proof heavily uses Khalil ('23+).

Proof of Theorem 3

- Let ν be the lift of μ to the graph of f in $\mathbb{R}^{k+d}$. Note $\widehat{\nu}(\mathbf{0}, \xi) = \widehat{\mu}(\xi)$.
- Consider copies of $\text{supp}(\mu)$ of size $\approx \delta$. Each has mass $\lesssim \delta^s$.
- Lift to graph. Push to tangent plane (FT are comparable up to error). FT of flattened piece is related to FT of μ at projected frequency.
- By non-degeneracy of f the projected frequencies are well separated, so by PFDA result most have Fourier decay.

Thank you for listening!